

DBMS ASSIGNMENT

Hotel Management System

Entity Relationship (ER) Diagram and Relational Schema

Application Domain: Hospitality / Hotel Management

1. Entities and Attributes

The Hotel Management System contains nine main entities covering guest management, room inventory, reservations, staff, payments, additional services, service orders, and feedback.

Entity	Attributes
Guests	GuestID (PK), FirstName, LastName, Email, Phone, Address, IDProofType, IDProofNumber, RegistrationDate
RoomTypes	RoomTypeID (PK), TypeName, BasePrice, MaxOccupancy, Description
Rooms	RoomID (PK), RoomNumber, RoomTypeID (FK), FloorNumber, Status
Staff	StaffID (PK), FirstName, LastName, Role, Phone, Email, HireDate
Reservations	ReservationID (PK), GuestID (FK), RoomID (FK), StaffID (FK), CheckInDate, CheckOutDate, NumGuests, Status, BookingDate
Payments	PaymentID (PK), ReservationID (FK), Amount, PaymentDate, PaymentMethod, PaymentStatus
Services	ServiceID (PK), ServiceName, Price, Description
ServiceOrders	ServiceOrderID (PK), ReservationID (FK), ServiceID (FK), Quantity, OrderDate, TotalAmount
Feedback	FeedbackID (PK), ReservationID (FK), GuestID (FK), Rating, Comments, FeedbackDate

2. ER Diagram

The following ER diagram shows the entities, attributes, primary keys, foreign keys, relationships, and cardinalities of the Hotel Management System.

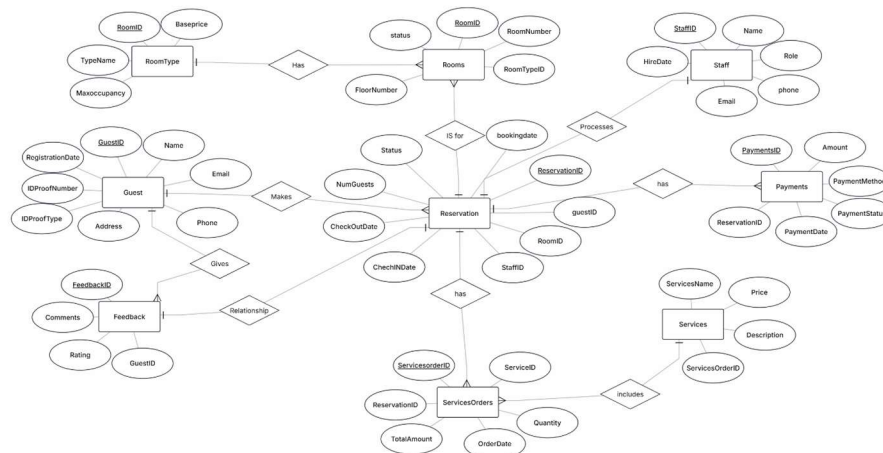


Figure 1: Entity Relationship Diagram – Hotel Management System

3. Relationships and Cardinality

Relationship	Cardinality
RoomTypes → Rooms	1 : M
Guests → Reservations	1 : M
Rooms → Reservations	1 : M
Staff → Reservations	1 : M
Reservations → Payments	1 : M
Reservations → ServiceOrders	1 : M
Services → ServiceOrders	1 : M
Reservations → Feedback	1 : 1
Guests → Feedback	1 : M
Reservations ↔ Services	M : N (resolved through ServiceOrders)

4. Relational Schema

- GUESTS(GuestID PK, FirstName, LastName, Email, Phone, Address, IDProofType, IDProofNumber, RegistrationDate)
- ROOMTYPES(RoomTypeID PK, TypeName, BasePrice, MaxOccupancy, Description)
- ROOMS(RoomID PK, RoomNumber, RoomTypeID FK, FloorNumber, Status)
- STAFF(StaffID PK, FirstName, LastName, Role, Phone, Email, HireDate)
- RESERVATIONS(ReservationID PK, GuestID FK, RoomID FK, StaffID FK, CheckInDate, CheckOutDate, NumGuests, Status, BookingDate)
- PAYMENTS(PaymentID PK, ReservationID FK, Amount, PaymentDate, PaymentMethod, PaymentStatus)
- SERVICES(ServiceID PK, ServiceName, Price, Description)
- SERVICEORDERS(ServiceOrderID PK, ReservationID FK, ServiceID FK, Quantity, OrderDate, TotalAmount)
- FEEDBACK(FeedbackID PK, ReservationID FK, GuestID FK, Rating, Comments, FeedbackDate)

5. Relational Schema Diagram

The following diagram presents the table structure and the PK/FK connections used in the database design.

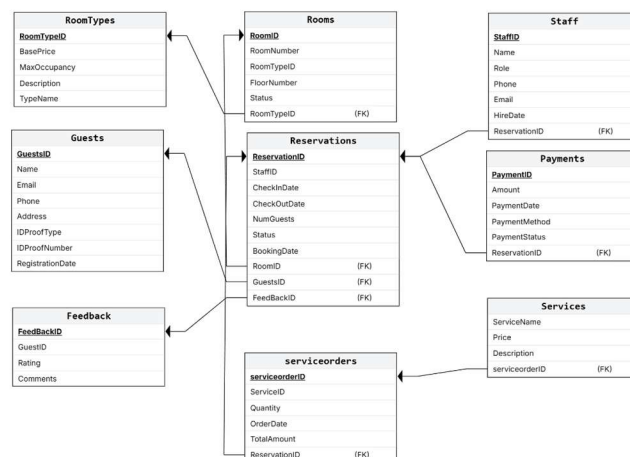


Figure 2: Relational Schema – Hotel Management System

6. Primary Key and Foreign Key Identification

Table	Primary Key (PK)	Foreign Key (FK)
Guests	GuestID	—
RoomTypes	RoomTypeID	—
Rooms	RoomID	RoomTypeID → RoomTypes
Staff	StaffID	—
Reservations	ReservationID	GuestID → Guests; RoomID → Rooms; StaffID → Staff
Payments	PaymentID	ReservationID → Reservations
Services	ServiceID	—
ServiceOrders	ServiceOrderID	ReservationID → Reservations; ServiceID → Services
Feedback	FeedbackID	ReservationID → Reservations; GuestID → Guests

7. Relationship Explanation

RoomTypes – Rooms (1:M): One room type can apply to many rooms, while each room belongs to one room type.

Guests – Reservations (1:M): One guest can make many reservations over time, while each reservation is made by one guest.

Rooms – Reservations (1:M): A room can appear in many reservations across different dates, while each reservation books one room.

Staff – Reservations (1:M): One staff member can process many reservations, while each reservation is processed by one staff member.

Reservations – Payments (1:M): One reservation can have multiple payments, such as advance and final settlement.

Reservations – Services (M:N): A reservation can include many services and a service can be used in many reservations. ServiceOrders resolves this many-to-many relationship.

Reservations – Feedback (1:1): Each reservation can have at most one feedback entry.

Guests – Feedback (1:M): A guest can provide feedback for multiple stays, while each feedback entry belongs to one guest.

8. Conclusion

The Hotel Management System database design represents the complete operational cycle of a hotel: guest registration, room categorisation and inventory, reservations, staff processing, payments, additional services, service orders, and guest feedback. Primary keys uniquely identify records, while foreign keys connect related tables and maintain data consistency.